19/08/2020

1. $\frac{1}{s(s+1/4)}$ $\quad F(s) = \frac{1}{s+1/4}$ = ~~[illegible]~~

$$L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t L^{-1}[F(s)]\,dt.$$

$$= \int_0^t L^{-1}\left(\frac{1}{s+1/4}\right]dt$$

$$= \int_0^t e^{-1/4 t}\,dt$$

$$\left[\frac{e^{-1/4 t}}{-1/4}\right]_0^t$$

$$-4\left[e^{-1/4 t} - e^0\right]$$

$$= -4\left[e^{-1/4 t} - 1\right].$$

2. $\dfrac{1}{s^2(s^2-9)}$

$F(s) = \dfrac{1}{s^2-9}$ $\quad L^{-1}[F(s)] = e^{9t}$

$$L^{-1}\left[\frac{F(s)}{s^2}\right] = \int_0^t\int_0^t L^{-1}[F(s)]\,dt$$

$$= \int_0^t\int_0^t e^{9t}\,dt$$

$$= \int_0^t \left[\frac{e^{9t}}{9}\right]_0^t dt.$$

$$= \frac{1}{9}\int_0^t [e^{9t}-1]\,dt$$

$$= \frac{1}{9}\left[\frac{e^{9t}}{9} - t\right]_0^t$$

$$= \frac{1}{9}\left(\frac{e^{9t}}{9} - t - \left[\frac{e^0}{9} - 0\right]\right)$$

$$= \frac{1}{9}\left[\frac{e^{9t}}{9} - t - \frac{1}{9}\right].$$

3. $\dfrac{5}{s^3-5s}$

$\dfrac{5}{s(s^2-5)}$ $\quad f(s) = \dfrac{5}{s^2-5}$

$$L^{-1}\left[\frac{f(s)}{s}\right] = \int_0^t L^{-1}[f(s)]\,dt$$

$$L^{-1}[f(s)] = \sqrt{5}\,\frac{\sqrt{5}}{s^2-5}$$

$$= \sqrt{5}\,\sinh\sqrt{5}\,t.$$

$$L^{-1}\left[\frac{f(s)}{s}\right] = \sqrt{5}\int_0^t \left[\frac{\cosh\sqrt{5}\,t}{\sqrt{5}}\right]_0^t$$

$\cosh\sqrt{5}t - \cosh 0$

$= \cosh\sqrt{5}t - 1$

4. $1/s^4 - 4s^2$

$$\frac{1}{s^2(s^2-4)} \qquad F(s) = \frac{1}{s^2-4} = \frac{1}{2}\,\frac{2}{s^2-4}$$

$$L^{-1}[F(s)] = \frac{1}{2}\sinh 2t.$$

$$L^{-1}\left[\frac{F(s)}{s^2}\right] = \frac{1}{2}\int_0^t\int_0^t \sinh 2t \, dt\, dt$$

$$= \frac{1}{2}\int_0^t \left[\frac{\cosh 2t}{2}\right]_0^t dt$$

$$= \frac{1}{4}\int_0^t [\cosh 2t - 1]\, dt$$

$$= \frac{1}{4}\left[\frac{\sinh 2t}{2} - t\right]_0^t$$

$$= \frac{1}{4}\left[\frac{\sinh 2t}{2} - t - \left[\frac{\sinh 0}{2} - 0\right]\right]$$

$$= \frac{1}{4}\left[\frac{\sinh 2t}{2} - t\right].$$

5. $\frac{2}{s(s^2+9)}$

$$F(s) = \frac{2}{s^2+9} = \frac{2}{3}\,\frac{3}{s^2+9}$$

$$L^{-1}[F(s)] = \frac{2}{3}\sin 3t.$$

$$L^{-1}\left[\frac{F(s)}{s}\right] = \frac{2}{3}\int_0^t \sin 3t\, dt$$

$$\frac{2}{3}\left[\frac{-\cos 3t}{3}\right]_0^t$$

$$\frac{2}{9}[-\cos 3t + 1]$$

1. $\frac{1-e^t}{t}$.

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds.$$

$f(t) = 1-e^t$

$F(s) = 1/s - \frac{1}{s-1}$

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \frac{1}{s} - \frac{1}{s-1}\,ds$$

$$= \left[\ln s - \ln s-1\right]_s^\infty$$

$$= \left[\ln \frac{s}{s-1}\right]_s^\infty$$

$$= \left[\ln \frac{1}{1-1/s}\right]_s^\infty$$

$$= \ln \frac{1}{1-1/s} - \ln \frac{1}{1-0}$$

$$= -\ln 1/1-1/s.$$

2. $\frac{2-2\cosh t}{t}$.

$$F(s) = \frac{2}{s} - \frac{2s}{s^2-1}$$

$$\left[2\ln s - \ln(s^2-1)\right]_s^\infty$$

$$\left[\ln \frac{s^2}{s^2-1}\right]_s^\infty$$

$$\left[\ln \frac{1}{1-1/s^2}\right]_s^\infty = -\ln 1/1-1/s$$

3 $\frac{2(1-\cos t)}{t}$

$f(t) = 2(1-\cos t)$

$F = 2\left(\frac{1}{s} - \frac{s}{s^2+1}\right)$

$= 2\left[\ln s - \frac{1}{2}\int_s^\infty \frac{2s}{s^2+1}\right)$

$2 = [\ln s]_s^\infty - \frac{1}{2}[\ln s^2+1]_s^\infty$

$2 = [\ln s - (\ln (s^2+1)^{1/2}]_s^\infty$

$= \left[\ln \frac{s}{(s^2+1)^{1/2}}\right]_s^\infty$

$= \left[\ln \frac{1}{\left(\frac{s^2+1}{s^2}\right)^{1/2}}\right]_s^\infty$

$= \left[\ln \frac{1}{\left[\frac{1}{s^2}+1\right]^{1/2}}\right]_s^\infty$

= ~~Indeterminate form~~

$= \ln \frac{s}{(1+1/s^2)^{1/2}}$

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